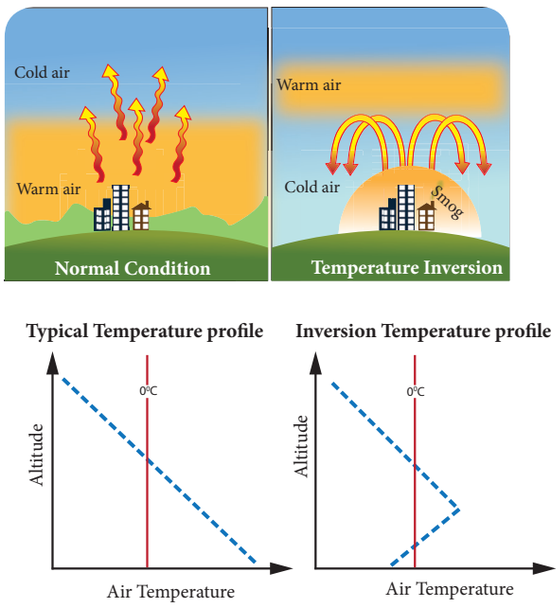



**Figure 6.7** Vertical distribution of Temperature

**Student Activity**

If the temperature of Chennai (7 m) is 34°C, calculate the temperature of Kodaikanal (2133m) using normal lapse rate.

### 6.3.3 Inversion of Temperature

The condition at which the temperature increases with altitude is called as 'inversion of temperature'. In this condition, warm air lies over cold air.

The conditions for inversion of temperature are:

a. **Long winter nights:** The bottom layer of the atmosphere in contact with the ground is cooled and the upper layer remains relatively warm.

b. **Cloudless sky:** The higher amount of terrestrial radiation reaches the higher altitude which leads to lower temperature at low level due to clear sky.

c. **Dry air near the surface:** the dry air absorbs less terrestrial radiation and allows them to escape into space.

d. **Snow covered ground:** During night, due to terrestrial radiation and higher albedo, most of the heat is lost to the atmosphere and the surface is cooled.

e. **Formation of fronts:** the movement of warm air over the cold air during the formation of the various fronts leads to inversion condition.

f. **Mountain wind:** The subsidence of cold mountain wind at the early morning leads to the displacement of warm air from the valley to higher altitude. This type of inversion is called as 'valley inversion'.




Albedo is the amount of solar radiation reflected from the surface. The variation is based on the nature of the earth's surface. Snow has higher albedo compared to forest.

### 6.3.4 Measurements of Temperature

| Unit of Temperature | Scientist | Year |
|---|---|---|
| Fahrenheit | Gabriel Fahrenheit | 1714 |
| Celsius | Andrew Celsius | 1742 |
| Kelvin | Lord Kelvin | 1848 |



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